

SAAKSHI · Camera coverage gap analysis

Gujarat Police Innovation Challenge 2026 - Pillar 1, Model 1 deliverable

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1 · Read this before any percentage

This report covers 50 of 80 registered cameras. The other 30 are unassessable, not uncovered: the upstream catalogue publishes {id, name} only, so they carry no coordinates and no district, and no spatial question can be asked of them at all. They are counted in camera_coverage - one row each, null geometry, with the reason recorded - precisely so that a later count(*) cannot be mistaken for the size of the estate.

80 cameras is a sandbox, not Gujarat. Every coverage cell is a circle: cameras has no bearing column, so a directional wedge is not expressible for any camera in the estate. The radii are assumptions, not survey measurements. No live VAHAN / SARTHI / eGujCop / AFIS / NAFIS connectivity is used anywhere, and no face recognition is performed.

2 · The headline

Every metre of the 20.35 km this estate appears to cover is contributed by a camera nobody has verified. Trusted-only coverage is 0.00 km - a delta of 20.35 km, or 100% of apparent coverage. The cause is measured and specific: 80 of 80 cameras have never had a health check run against them, so every one resolves to band: null. That is an absence of evidence, not a bad result - but a conventional coverage map would have drawn all 20.35 km of it in green.

slice	cameras	covered	share of network
All cameras	50	20.35 km	0.0093%
Trusted cameras only	0	0.00 km	0.0000%
ANPR-viable cameras only	17	2.77 km	0.0013%

Mapping where cameras are is a map anyone can draw. cam03 returned 67 sightings in the same city and hour that cam04 returned 33,548 - a 500x spread - and cam03 is not broken: it decoded 5,582 frames cleanly at 23.16 fps. It simply sees almost no vehicles (D1-09, #13). A coverage polygon for cam03 would be the same size and colour as cam04's. Yield is not a geometric property, which is why the trust band, not the polygon, decides whether a metre of road counts as covered. Likewise 8 of 30 measured cameras fail night usability (D1-05, #9): a junction covered only by a night-blind camera is uncovered for half of every day.

3 · Method

Cell radius by geometry class, all assumed: anpr_viable 60 m, detection_only 120 m, unclassified 60 m. Denominator: 218137.50 km over 5,40,711 ways; service, track, path, footway and cycleway were excluded at import (D3-01, #24). ST_Intersection and ST_Difference are evaluated independently in EPSG:32643 and reconciled to 1 m; the generator refuses to publish if a slice falls outside that.

slice	cand. ways	cand. km	covered	reconcile err
All cameras	261	77.04 km	20.35 km	0.000000 m
Trusted cameras only	0	0.00 km	0.00 km	0.000000 m
ANPR-viable cameras only	43	15.61 km	2.77 km	0.000000 m

4 · Junctions with zero trusted coverage

A junction is a point where at least 3 distinct ways of class motorway, trunk, primary, secondary, tertiary terminate - a topological definition from way endpoints, not an OSM junction tag, so it misses junctions where one way passes through another without a shared endpoint. 6,750 of 6,750 have no trusted camera covering them.

#	lon	lat	ways nearest named way
1	72.570734	23.186064	6 Amba Township Main Road
2	73.787246	20.018091	6 -
3	68.977376	22.248284	5 Station Road
4	69.606003	21.640595	5 MG Road
5	70.459215	21.739962	5 -
6	70.617416	21.771482	5 -
7	70.829410	22.819765	5 -
8	70.848631	22.825083	5 -
9	71.352959	24.154854	5 -
10	71.457471	23.011033	5 -
11	71.493096	22.998067	5 State Highway 7
12	71.628397	22.719469	5 -
13	72.064622	23.093275	5 -
14	72.137592	23.851330	5 -
15	72.337200	23.288769	5 Kadi Underpass
16	72.384048	23.567135	5 Mehsana-Himatnagar Vijapur road
17	72.390459	23.603880	5 Station Road
18	72.488464	23.241196	5 -
19	72.507990	23.050416	5 Thaltej Road
20	72.511614	23.027296	5 Ramdevnagar road
21	72.562200	23.009330	5 Ashram Road

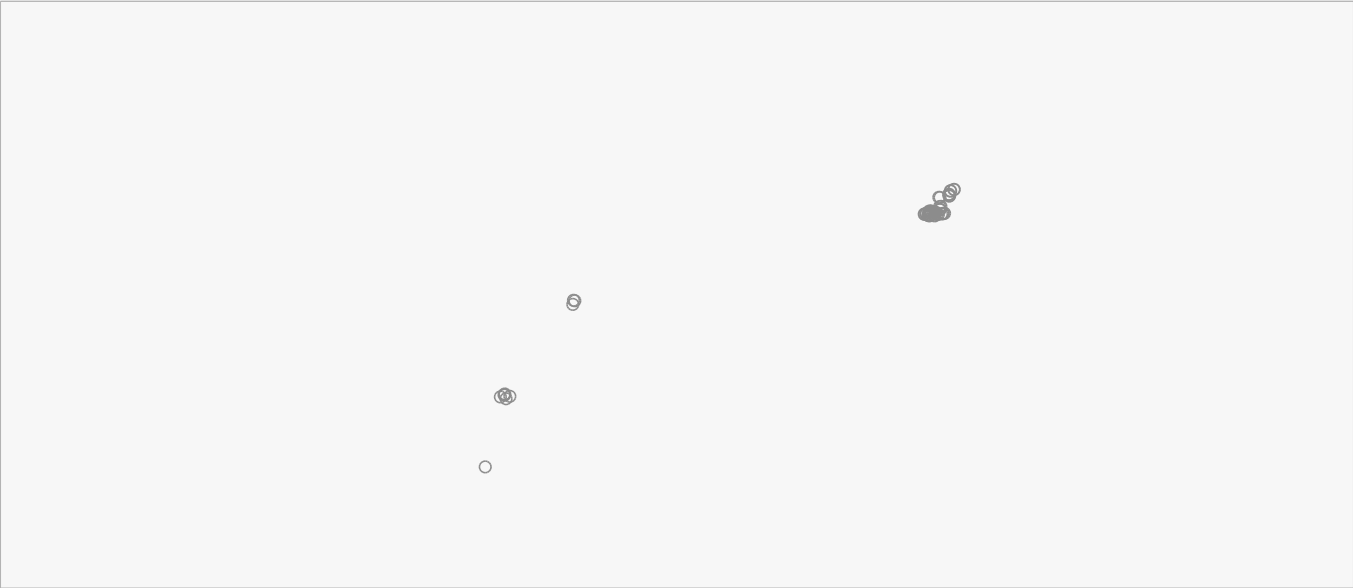
#	lon	lat	ways	nearest named way
22	72.570750	23.042429	5	Gujrat vidhyapeeth road
23	72.571280	23.022565	5	Ashram Road
24	72.575148	23.123451	5	-
25	72.596469	22.996112	5	-

5 • Where the coverage is

district	covered km (all cameras)
Junagadh	0.81 km
Gandhinagar	3.82 km
Rajkot	1.71 km
Ahmedabad	12.99 km
Gir Somnath	1.03 km

No per-department breakdown is given, and that is itself a finding: department_id is NULL for every camera in this estate, so a departmental trust deficit has exactly one bucket - no owning department recorded (D3-05, #28).

Plan view



50 placed cameras of 80 registered, across 5 districts. Frame is the import clip 68,19.9 to 74.6,24.8. Dark rings counted as trusted (0); light rings did not. Markers show position only - a 60 m cell is far below one point at this scale, so the rings are not drawn to scale and this figure must not be read as a coverage area. The 30 cameras with no coordinates cannot appear here at all.

6 • Figures this run did not measure

Measured against the live sandbox feed by earlier tickets. The feed is not held during report generation, so these are cited, not recomputed, and no figure above depends on them.

fact	value	source
Per-camera sighting yield, same city, same hour	a 500x spread - cam04 33,548 · cam08 24,462 · cam05 17,622 · c...	D1-09 (#13)
cam03 is not broken	decoded 5,582 frames cleanly at 23.16 fps - it simply sees almost no...	D1-09 (#13)
Cameras failing night usability	8 of 30; luma means range 8.40 to 135.19	D1-05 (#9)
Distinct resolutions across the measured estate	six - 854x480 x12 · 1920x1080 x11 · 1280x960 x3 · 1280x720 x2 · ...	D1-05 (#9)
Cameras effectively blind on focus (quality 0, a dis...	cam22 (blur 0.011) and cam09 (blur 2.047)	D1-05/D1-06 (#9, #10)
Cameras declaring a retention period	0 of 30	D3-05 (#28)
Road network	540,584 GiST-indexed ways from Geofabrik western-zone, clipped to...	D3-01 (#24)